Prodigy Anesthesia Quick Review

Sepsis

What is septic shock?	Septic shock is a type of distributive shock that occurs due to a widespread inflammatory response. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1589.*
What organ system is most susceptible to ischemia from septic shock?	Because of the redistribution of blood that occurs in septic shock, the gastrointestinal tract is the most susceptible to ischemia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1589.*
How does septic shock affect acid-base balance?	Because of an increased metabolic demand and an inability to perfuse tissues adequately, there is often a metabolic acidosis present. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1589-1590.*
How is cardiac contractility affected by septic shock?	There is an overall decrease in cardiac contractility. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1589.*
What are the cardiovascular changes associated with septic shock?	High cardiac output, low systemic vascular resistance, hypotension, and tissue underperfusion. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1589.*
A patient in septic shock has a pH of 7.2. Should sodium bicarbonate be administered?	No. Patients in septic shock with a pH higher than 7.15 should not be given sodium bicarbonate. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1591.*
What is the indication for the administration of recombinant activated protein C?	Recombinant activated protein C should be administered if the patient is at a high risk of death. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1593.*
When should steroids be used in the treatment of sepsis?	Steroids may be administered in patients whose blood pressure does not respond adequately to fluid resuscitation and pressors. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1591.*

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How do the roles of norepinephrine, dopamine, vasopressin, and dobutamine compare in the treatment of sepsis?	Norepinephrine and dopamine are considered the first-line treatments in sepsis to achieve a mean arterial pressure of 65 mmHg or greater. Vasopressin may be used at a fixed rate as an adjunct to norepinephrine and dopamine if needed. Low-dose dopamine for 'renal protection' is not recommended for these patients. If the patient exhibits a low cardiac output despite fluid resuscitation, dobutamine may be used. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1591.*
What ventilator settings are most appropriate for the patient in septic shock?	The current recommended protocol is a low tidal volume, low inspiratory pressure, and the application of PEEP. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1591.*

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